Project 2

Due: Apr. 1st, 11:59 pm

Obtain a numerical solution to the zero pressure gradient laminar boundary layer equations:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \frac{\partial}{\partial y}\left(\nu\frac{\partial u}{\partial y}\right)$$

Use the Crank-Nicolson algorithm to march in the *x* direction. Use a stretched grid (see the hint on the next page) in the *y* direction and a fourth-order central difference scheme for discretization in η, except at the first interior points where a second order discretization should be used. η is the stretched coordinate that is uniform in the computational domain.

Use the following scales:

$$\nu_\infty = 1.5 \times 10^{-6}\frac{\mathrm{m}^2}{\mathrm{s}}, U_\infty = 30.0\frac{\mathrm{m}}{\mathrm{s}}, L = 1.0\ \mathrm{m}, \delta = 0.005\ \mathrm{m}$$

At the initial location, assume that the velocity profile in the x direction is given by

$$\mathrm{u/U_\infty} = \begin{cases} \frac{y}{\delta}\left(\frac{3}{2} - \frac{1}{2}\left(\frac{y}{\delta}\right)^2\right), \text{for } \frac{y}{\delta} \le 1 \\ 1, \text{for } \frac{y}{\delta} > 1 \end{cases}$$

Use 150 grid points in the *y* direction with $y_{max} = 10\delta$.

Provide velocity profiles at four axial locations (as well as the initial location) up to 40 *L* downstream of the initial location. Estimate the initial value of *x* based on the growth of the boundary layer. Compare your results with Blasius solution at the subsequent downstream locations. Calculate the displacement and momentum thicknesses and the shape factor in the *y* direction.

Other requirements:

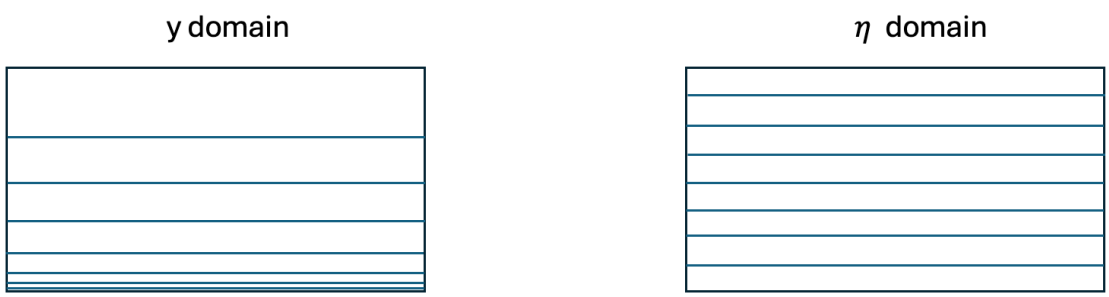
1. The report should not exceed 5 pages. Appendices should include details of your computer code.
2. Preferred programming language for this project is FORTRAN or C or C++. Please consult the instructor if you plan to use other languages.
3. A modular programming practice is highly recommended.

Hints:

The following form of a stretch grid is recommended, where the physical coordinate y is related to the numerical domain coordinate η as

$$y = y_{max} \frac{\sinh(s\eta)}{\sinh(s)}$$

where s is a stretching factor. The reason of using the stretch grid is to simplify discretization because uniform grid is used in η, while physically having more refined grid near the wall to resolve the velocity gradient near the wall. A smaller value of s results in almost linear relations between y and η, while a larger value of s results in a more wall-concentrated grid in the physical y domain. A value of s = 10 is recommended.



The governing equation needs to be converted into the η coordinate, using the chain rule, as follows,

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial y}$$

$$\frac{\partial^2}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial}{\partial \eta}\frac{\partial \eta}{\partial y}\right)$$

ME 5311: Computer Project #2: Grade Sheet

Item	Score
Description of the problem	/01
Description of the numerical method	/03
Presentation of results	
Velocity profiles	/07
Calculation of initial value of x	/01
Comparison with Blasius solution	/02
Calculation of thicknesses and shape factor	/02
Discussion of results	
General description	/01
Accuracy and stability	/01
Appendix – copy of program listing	
Structure	/01
Documentation	/01
Total:	/20